



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam Fall 2024

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: 40

Duration: 2 Hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Consider the following **Context-free grammars (CFG)** and answer according to it:

[4*2]

- a) $S \rightarrow XYZ$
 $X \rightarrow 0X \mid 1X \mid \varepsilon$
 $Y \rightarrow A \mid B$
 $Z \rightarrow 0Z1 \mid Z0 \mid Z1 \mid \varepsilon$
 $A \rightarrow aA \mid a$
 $B \rightarrow Bb \mid bB \mid b$

With the help of **Top-Down Parse Trees**, find-out if the grammar is **Ambiguous** or not for the string: **“011b01”**

- b) $S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid E \mid \text{other}$
 $E \rightarrow E \ \&\& \ E \mid E \parallel E \mid !E \mid (E) \mid \text{id}$

With the help of the **rightmost derivation**, derive the following string **“if id $\parallel$ id $\&\&$ id then if id then other else other”**

2. Design **CFGs that generate** the following languages for $\Sigma = \{a, b\}$:

[2*4]

- a) $L = \{ a^n b^m \mid n \leq m \leq 2n \text{ and } n, m \geq 1 \}$
b) $L = \{ a^m b^{2n} c^{3n} d^{3m} \mid \text{where } n \geq 2, m \geq 1 \}$
c) $L = \{ a^i b^j c^k \mid \text{where } i=k \text{ or } j=k \text{ and } i, j, k > 0 \}$
d) $L = \{ \text{every even length palindrome when the string starts with a or every odd length palindrome when the string starts with b} \}$

3. Convert the following grammars into **Chomsky Normal Form (CNF)**:

[4*2]

- a) $S \rightarrow S+S \mid T-S \mid S/T \mid \varepsilon$
 $T \rightarrow T*TC \mid C$
 $C \rightarrow 0 \mid 1 \mid 2 \mid 3$
b) $S \rightarrow SS \mid CAB \mid BA$
 $A \rightarrow aAb \mid \varepsilon$
 $B \rightarrow 0A1 \mid 0B1$
 $C \rightarrow cA \mid Bb$

4. Draw the **Push Down Automata (PDA)** for the following languages:

[4*2]

- a) $L = \{ x^{3a} y^{2b} z^c w^d \mid (c = 3b \text{ and } a=2d) \text{ and } a \geq 1, b \geq 0 \}$
- b) $L = \{ p^{3i} \# q^{2j} r^{2k} \mid (i=j \text{ or } i=k) \text{ and } i, j \geq 1 \}$

5. Draw a **Turing Machine** for the following language and show the **Tape Traversal** to validate the given input: [4*2]

$$L = \{ a^p b^{3q} c^r d^x \mid r = 2p + q \text{ and } q = x - p \text{ and } p, q, r, x \geq 1 \}$$

Input String: **aabbbccccdd**